

Project Proposal

Proper waste management is a crucial step in reducing the environmental impacts of a growing population that produces an excess amount of waste. Having a means of sorting waste correctly and efficiently ensures that these materials are processed properly, which in turn will minimize environmental pollution. We propose a convolutional neural network (CNN) as a solution to learn various categories of waste, that can be used by recycling plants to automatically detect the category of waste via live video and effectively sort it. We intend to first leverage You Only Look Once (YOLO) to detect objects in a frame, then use our trained CNN with transfer learning (can try ResNeXt-101, ResNeXt-50, or MobileNetV3-Large models) to classify the image, which is an appropriate deep learning network for detecting spatial patterns.

The datasets we intend to use include Trash Annotations in Context (TACO)^[1], Trashbox^[2], and Trashnet^[3]. TACO has 1,500 samples of real-world images with complex backgrounds, labelled with around 60 imbalanced classes that we can select from and consolidate into broader classes. Trashbox has 14,282 samples of 7 classes (1,930 cardboard, 2,156 paper, 2,135 plastic, 2,022 glass, 2,068 metal, 2,406 e-waste, and 1,565 medical waste images) and Trashnet has 2,527 samples of 6 classes (403 cardboard, 501 glass, 410 metal, 594 paper, 482 plastic, and 137 trash images). In total, there is potential to use 18,309 samples. As a pre-processing step, we intend to create augmented images to help with model generalization, involving translations, rotations, scaling, and background switching. To get new samples, we each intend to collect images of trash through our webcams, since the intent of this model is to be able to identify images via webcams in a sorting facility. We also intend to each take pictures of trash in our houses that can be added into the model training set.

To demonstrate the capabilities of the model, we will use three test sets, one being a holdout set from the original datasets that we obtain, we will showcase how our model works on new test images that we take, and lastly, we will deploy the model to integrate with a live video stream containing trash in the field of view. We will use metrics such as accuracy, precision, recall, F1 score, and a confusion matrix to understand what is being misclassified. This project will be successful because the categories that we will be classifying the images into have strong visual patterns (distinctive colours and textures), plus there have been existing research that was deemed successful while also using the datasets that we intend to use. Lastly, combining models such as YOLO with transfer learning models should lead to a robust and efficient classification process with real-world applications. Real-world impacts of a successful implementation of this model could lead to higher recycling rates, reduced human interference/labour in waste sorting, make waste management more accessible and sustainable, and ultimately yet most importantly reduce the environmental impact of waste, including pollution, landfill use, and resource depletion.

^[1] <http://tacodataset.org/>

^[2] <https://www.kaggle.com/datasets/minhle13/trashbox>

^[3] <https://www.kaggle.com/datasets/feyzazkefe/trashnet>